

Arselene Belhayej Software Engineer

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EDUCATION

Master of Science in IT Infrastructure and Cloud Technology
Private Higher School of Engineering and Technology

Sep 2022 – Oct 2025 | Ariana, Tunisia

B.S. in Computer Science & Business Intelligence
Higher Institute of Computer Science of Mahdia

Sep 2019 – Jul 2022 | Mahdia, Tunisia

WORKING EXPERIENCE

Software Engineer
IT SERV

Feb 2025 – Aug 2025 | Tunis, Tunisia

- Built and operated Spring Boot + Angular microservices, introducing Kafka, handling ~8,000 events.
- Deployed a fine-tuned TinyLlama 1.1B via FastAPI, added Prometheus/Grafana and ELK Stack logging, with Jenkins, Docker, Helm to Kubernetes deployments for enhanced reliability.

Technologies: Spring Boot, Angular, Kafka, Jenkins, Kubernetes, Docker, Helm, Grafana, ELK Stack, PostgreSQL

DevOps Intern

Jun 2024 – Sep 2024 | Zurich, Switzerland

SILVER BRAIN AI AG

- Delivered a cloud-based IDE using AWS Cloud9, Lambda, and API Gateway to speed up environment setup for the team.
- Improved CI/CD pipelines with Jenkins, Terraform, Kubernetes, Docker, reducing pipeline runtime to 14 minutes.

Technologies: AWS Cloud9, Lambda, API Gateway, Kubernetes, Terraform, Jenkins, Docker, Prometheus

Cloud Software Engineer

Sep 2023 – Feb 2024 | Ariana, Tunisia

EmeralDevs

- Operated AWS EC2 with Auto Scaling and S3 infrastructure to support dynamic software workloads, achieving 99.9% availability.
- Strengthened access with IAM policies via Azure SSO, Kubernetes RBAC, Docker, Terraform, and Jenkins for fortified security.

Technologies: AWS EC2, S3, Auto Scaling, IAM, Azure, Kubernetes, Docker, Terraform, Jenkins

Software Engineer Intern

Jun 2023 – Sep 2023 | Manouba, Tunisia

DataDoIT

- Orchestrated a deployment pipeline for MERN apps using Azure DevOps and Docker, cutting manual steps to 0.
- Built Grafana dashboards with Prometheus/Kubernetes metrics for real-time system monitoring, improving MTTD by ~50%.

Technologies: Azure DevOps, Terraform, Docker, Kubernetes, Prometheus, Grafana, Jenkins

Full-Stack Developer

Feb 2022 – Aug 2022 | Tunis, Tunisia

AKRUSS-IT

- Shipped an e-learning platform using React, Node.js/Express, MongoDB, and TypeScript for better user engagement.
- Analyzed data trends with Python and orchestrated deployment on AWS EC2 via Docker, reducing downtime by 40%.

Technologies: AWS EC2, TypeScript, Docker, React.js, Python, Express.js, MongoDB

PROJECTS

Automated Fitness Platform

Spring Boot, Angular, Docker, Kubernetes, Jenkins, Kafka, Keycloak, Python, ELK Stack

- Developed a microservices-based fitness app with Spring Boot, Angular, Keycloak, and Kafka for event streaming.
- Fine-tuned TinyLlama chatbot, automated CI/CD via Jenkins to Kubernetes with Grafana monitoring and ELK logging.

AI-Powered Auto-Healing and Predictive Scaling in Kubernetes

Kubernetes, TensorFlow, AWS EKS

- Trained TensorFlow models on Prometheus signals to predict scaling, reducing compute costs by 22%.
- Implemented automated recovery actions (pod restarts, node draining) on AWS EKS through Terraform and RBAC, cutting downtime by ~90 minutes/month.

AWS Infrastructure Setup

AWS, VPC, Subnets, NAT Gateway, ALB, Auto Scaling, Security Groups, S3

- Designed scalable and secure AWS architecture with VPC, subnets, and Auto Scaling to support high-traffic applications.
- Established advanced security measures using Security Groups and bastion hosts for precise access control.

SaaS Platform

Java, OpenStack, Node.js, Angular, Docker, Kubernetes, Heat, RabbitMQ

- Engineered a full-stack SaaS platform in Java and Angular, integrating RabbitMQ for async messaging, reducing workflow latency by 35%.
- Implemented CI/CD pipelines on Azure DevOps with Docker, Kubernetes, and Trivy, blocking 12 critical vulnerabilities within the images.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C#, SQL, Bash

Cloud & Infra: AWS, Azure, Kubernetes, Docker, Terraform, Ansible, Jenkins, Helm, Prometheus, Grafana, ELK, Kafka, Linux, KVM

Web/Backend: Spring Boot, Node.js/Express, FastAPI, React, Next.js, Angular, PostgreSQL, MongoDB, Linux, Git